

Adaptive Timing Scheduler for Low-Latency nFAPI Networks

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Abstract Section (before Introduction) Add a clear placeholder for an abstract:

Purpose: Briefly summarize the problem, method, results, and contribution.

Word count: typically 150—250 words.

Keywords: list 3—6 keywords.

I. Introduction

A. Introduction

Purpose: To explain what the paper is about, why the problem matters, and what the others authors have done to address it.

Typical content:

- Background and motivation
- Problem statement
- Goals of the paper
- Contributions (what's new or important)
- High-level overview of the approach or results

Example phrases:

- “In recent years, ...”
- “This paper addresses the problem of ...”
- “Our main contributions are ...”

B. Related Works

Purpose: To show how your work fits into the existing body of knowledge, and how it differs from or improves upon previous work.

Typical content:

- Summary of key prior research
- Strengths and limitations of existing approaches
- Comparison to your approach
- Justification for why a new approach is needed

Example phrases:

- “Several studies have explored ...”
- “Unlike [Author], we propose ...”
- “Prior work has not considered ...”

The main contributions of this paper are summarized as follows.

- Contribution 1.
- Contribution 2.
- Contribution 3.

The rest of the paper is organized as follows. Section II defines the system model/architecture considered in this paper. The proposed analytical model/method is given in Section III. Section IV shows the numerical results. Concluding remarks are given in Section VI.

II. System Model/Architecture

TABLE I: nFAPI Delay Management Configuration Parameters

Parameter	Type	Description
timingWindow	uint16_t	Timing window for delay management. See section 3.2.9 of [10]
timingMode	uint8_t	Timing mode for Timing Info. See section 3.2.9 of [10]
timingPeriod	uint8_t	Timing period for Timing Info. See section 3.2.9 of [10]

Figure ?? shows the system model/architecture considered in this paper. (Draw a figure showing the key components and their relationship in the system model. It may consist the I/P and O/P parameters of your system or the major components showing your contributions.)

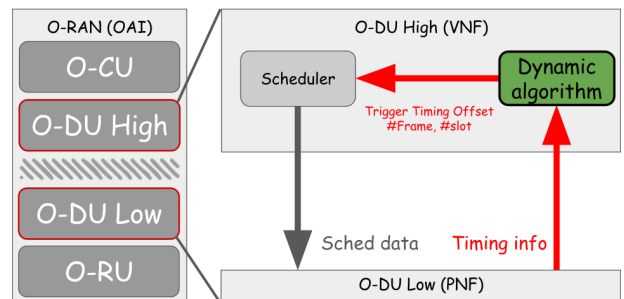


Fig. 1. System architecture of the ORAN based implementation.

The following assumptions were made in this paper.
 TODO: Add explanatory text for each assumption.

$$\hat{J}_k = \alpha \hat{J}_{k-1} + (1 - \alpha) \Delta t_k,$$

where $0 < \alpha < 1$ is a smoothing factor. The desired offset δ that aligns the scheduler transmission with the PHY deadline is

$$\delta_k = T_{\text{deadline}} - (\Delta t_k + \hat{J}_k).$$

The offset is then converted to frame and slot indices:

$$f_k = \left\lfloor \frac{\delta_k}{T_{\text{frame}}} \right\rfloor, \quad s_k = \left\lfloor \frac{\delta_k - f_k T_{\text{frame}}}{T_{\text{slot}}} \right\rfloor,$$

where $T_{\text{textframe}} = N_{\text{textframe}} T_{\text{textslot}}$.

The scheduler applies (f_k, s_k) to the scheduling request such that the transmitted data arrives at the ODU Low exactly at the required PHY boundary.

Fig. 2. Threestage closedloop timing adjustment.

- Assumption 1.
- Assumption 2.
- Assumption 3.

TABLE II
Parameter Summary (TODO)

Parameter	Value
T_{slot}	0.5 ms
N_{frame}	10
α	TODO

III. Proposed Method

In this work we propose a closedloop timing adjustment mechanism that integrates the dynamic algorithm described in Section IIC with the ODU scheduler. The method consists of three stages:

1. **Timing Feedback Collection** — ODU Low measures the actual transmission time of each downlink subframe and reports a timing offset Δt_k to the ODU High via the FAPI interface.
2. **Offset Computation** — The dynamic algorithm at ODU High processes Δt_k using the EMA model above and computes the trigger timing offset (f, s) .
3. **Scheduler Adaptation** — The computed offset is injected into the MAC scheduler's timing parameters so that the next scheduling request is shifted accordingly.

The algorithm operates on a per-slot basis, ensuring that the control loop latency remains below one subframe (≤ 0.5 ms). By continuously adapting to the measured jitter, the system can compensate for virtualization-induced delays and maintain tight synchronization between the VNF and the hardware-accelerated PHY.

IV. Numerical/Simulation/Experimental Results

Computer simulations/Experiments were conducted to verify the effectiveness of the proposed analytical model.

Fig. 3. Validation workflow for the proposed method.

Fig. 4. Jitter reduction over iterations (TODO: add data).

Fig. 5. Trigger offset (f, s) versus measured delay (TODO).

Fig. 6. Throughput comparison with baseline (TODO).

XXX scenarios were investigated. Scenario I was designed to demonstrate/verify the accuracy/effectiveness of Contribution 1.

- A. Contribution 1
- B. Contribution 2
- C. Contribution 3

References

- [1]

TABLE III
Simulation parameters for each scenario.

Scenario	Parameter	Value
Scenario 1	VNF CPU cores	TODO
	Feedback period	TODO
Scenario 2	PHY latency	TODO
	EMA α	TODO

Fig. 7. Result for Contribution 1 (TODO).

Fig. 8. Result for Contribution 2 (TODO).

Fig. 9. Result for Contribution 3 (TODO).